

## ARVAIKHEER, Mongolia

WMO: 442880

Lat: 46.2592N

Lon: 102.7889E

Elev: 1813

StdP: 81.36

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |       |      |       | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|-------|------|-------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |       | 1%   |       |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB  | WS   | MCDB  | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)   | (l)  | (m)   | (n)                   | (o)  | (p)   |
| 1             | -25.5      | -23.6 | -32.3                         | 0.2 | -22.0 | -30.3 | 0.3 | -20.3 | 13.5                  | -15.6 | 11.7 | -14.6 | 2.3                   | 320  | 0.702 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 9.3                    | 27.6            | 14.6 | 25.8 | 13.9 | 24.1 | 13.4 | 17.0                | 24.5 | 15.8 | 22.7 | 14.8 | 21.4 | 4.5                  | 180  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 14.4                            | 12.8 | 19.6 | 13.3 | 11.9 | 18.5 | 12.2 | 11.1 | 18.0 | 55.2          | 24.3 | 51.3 | 22.8 | 48.0 | 21.3 | 23.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |  |
|-------------------|------|-----|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|--|
| 1%                | 2.5% | 5%  | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |  |
|                   |      |     | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |  |
| (a)               | (b)  | (c) | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |  |
| 12.7              | 10.5 | 9.7 | DB                         | -27.8 | 30.6               | 2.7 | 1.8                                                | -29.8 | 31.9       | -31.4 | 33.0       | -32.9 | 34.0       | -34.9 | 35.3 |  |
|                   |      |     | WB                         | -27.9 | 18.3               | 2.7 | 2.1                                                | -29.9 | 19.8       | -31.5 | 21.0       | -33.0 | 22.2       | -35.0 | 23.7 |  |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |       |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |       |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 2.4   | -14.3 | -11.3 | -4.4  | 3.8   | 9.8   | 15.4  | 17.9  | 16.0  | 10.5  | 2.7   | -6.3  | -12.0 | (6)  |
| (7)  |                                                                                  | DBStd       | 12.28 | 5.21  | 5.65  | 6.09  | 5.49  | 5.27  | 3.86  | 3.35  | 3.37  | 4.68  | 5.19  | 5.85  | 4.97  | (7)  |
| (8)  |                                                                                  | HDD10.0     | 3516  | 752   | 596   | 446   | 195   | 69    | 5     | 0     | 2     | 49    | 231   | 489   | 683   | (8)  |
| (9)  |                                                                                  | HDD18.3     | 5887  | 1011  | 829   | 704   | 437   | 266   | 102   | 49    | 87    | 236   | 486   | 739   | 941   | (9)  |
| (10) |                                                                                  | CDD10.0     | 738   | 0     | 0     | 0     | 8     | 64    | 168   | 244   | 188   | 64    | 3     | 0     | 0     | (10) |
| (11) |                                                                                  | CDD18.3     | 68    | 0     | 0     | 0     | 0     | 3     | 14    | 35    | 15    | 1     | 0     | 0     | 0     | (11) |
| (12) |                                                                                  | CDH23.3     | 539   | 0     | 0     | 0     | 0     | 24    | 140   | 253   | 112   | 9     | 0     | 0     | 0     | (12) |
| (13) |                                                                                  | CDH26.7     | 99    | 0     | 0     | 0     | 0     | 1     | 24    | 57    | 16    | 0     | 0     | 0     | 0     | (13) |
| (14) | Wind                                                                             | WSAvg       | 3.7   | 2.7   | 3.2   | 4.2   | 5.0   | 5.0   | 4.3   | 3.4   | 3.3   | 3.7   | 3.7   | 3.2   | 2.7   | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 234   | 1     | 1     | 4     | 7     | 15    | 35    | 81    | 61    | 18    | 5     | 3     | 2     | (15) |
| (16) |                                                                                  | PrecMax     | 398   | 10    | 7     | 18    | 30    | 56    | 130   | 181   | 152   | 71    | 27    | 13    | 11    | (16) |
| (17) |                                                                                  | PrecMin     | 100   | 0     | 0     | 0     | 0     | 0     | 1     | 5     | 5     | 0     | 0     | 0     | 0     | (17) |
| (18) |                                                                                  | PrecStd     | 71    | 2     | 2     | 4     | 7     | 13    | 22    | 44    | 35    | 15    | 7     | 3     | 2     | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 1.5   | 4.1   | 13.4  | 20.2  | 26.3  | 29.2  | 31.1  | 28.7  | 24.8  | 17.6  | 10.4  | 2.2   | (19) |
| (20) |                                                                                  | MCWB        | -3.2  | -1.3  | 3.9   | 8.1   | 10.9  | 14.3  | 17.4  | 15.6  | 12.5  | 6.8   | 2.8   | -3.0  | (20)  |      |
| (21) |                                                                                  | 2%          | DB    | -1.8  | 1.3   | 9.9   | 17.3  | 23.4  | 26.8  | 28.1  | 26.2  | 22.1  | 14.9  | 7.0   | -0.6  | (21) |
| (22) |                                                                                  | MCWB        | -5.4  | -3.4  | 1.8   | 6.4   | 10.2  | 13.6  | 15.7  | 14.4  | 10.9  | 5.8   | 0.7   | -4.4  | (22)  |      |
| (23) |                                                                                  | 5%          | DB    | -4.4  | -0.8  | 7.3   | 15.0  | 21.0  | 24.7  | 26.1  | 24.2  | 20.1  | 12.7  | 4.4   | -2.8  | (23) |
| (24) |                                                                                  | MCWB        | -7.4  | -4.8  | 0.3   | 5.3   | 9.1   | 12.7  | 14.9  | 13.9  | 10.1  | 4.6   | -0.9  | -6.1  | (24)  |      |
| (25) |                                                                                  | 10%         | DB    | -6.6  | -2.9  | 4.7   | 12.7  | 18.9  | 22.8  | 24.2  | 22.3  | 18.1  | 10.6  | 2.1   | -4.8  | (25) |
| (26) |                                                                                  | MCWB        | -9.1  | -6.3  | -1.2  | 4.2   | 8.1   | 12.1  | 14.4  | 13.1  | 9.2   | 3.4   | -2.4  | -7.6  | (26)  |      |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | -2.9  | -1.0  | 4.7   | 9.3   | 13.7  | 16.5  | 19.3  | 17.8  | 14.2  | 8.5   | 3.0   | -2.6  | (27) |
| (28) |                                                                                  | MCDWB       | 0.5   | 3.8   | 11.6  | 18.1  | 21.7  | 25.5  | 27.5  | 25.3  | 21.7  | 15.4  | 9.7   | 1.6   | (28)  |      |
| (29) |                                                                                  | 2%          | WB    | -5.2  | -3.2  | 2.2   | 7.1   | 11.6  | 15.1  | 17.6  | 16.3  | 12.3  | 6.4   | 0.9   | -4.4  | (29) |
| (30) |                                                                                  | MCDWB       | -1.7  | 0.8   | 9.2   | 15.4  | 20.2  | 23.5  | 25.3  | 23.3  | 19.2  | 13.8  | 6.6   | -0.7  | (30)  |      |
| (31) |                                                                                  | 5%          | WB    | -7.3  | -4.7  | 0.5   | 5.7   | 10.1  | 14.1  | 16.5  | 15.2  | 11.1  | 5.0   | -0.7  | -6.0  | (31) |
| (32) |                                                                                  | MCDWB       | -4.5  | -1.1  | 6.8   | 14.0  | 19.2  | 21.5  | 23.5  | 21.6  | 18.4  | 11.7  | 4.1   | -3.0  | (32)  |      |
| (33) |                                                                                  | 10%         | WB    | -9.1  | -6.2  | -1.0  | 4.4   | 8.8   | 13.2  | 15.4  | 14.2  | 9.9   | 3.7   | -2.3  | -7.6  | (33) |
| (34) |                                                                                  | MCDWB       | -6.6  | -3.0  | 4.5   | 12.1  | 17.7  | 20.2  | 21.8  | 20.2  | 17.1  | 10.2  | 1.9   | -4.9  | (34)  |      |
| (35) | Mean Daily<br>Temperature<br>Range                                               | MDBR        | 8.7   | 9.7   | 10.6  | 10.9  | 10.8  | 10.3  | 9.3   | 9.6   | 10.2  | 10.1  | 9.3   | 8.4   | (35)  |      |
| (36) |                                                                                  | 5% DB       | MCDBR | 11.3  | 11.9  | 13.0  | 13.3  | 12.9  | 12.7  | 12.0  | 12.1  | 12.1  | 11.2  | 10.8  | (36)  |      |
| (37) |                                                                                  | MCWBR       | 8.7   | 8.8   | 8.1   | 7.5   | 6.5   | 5.7   | 5.0   | 5.4   | 6.0   | 7.0   | 7.5   | 8.3   | (37)  |      |
| (38) |                                                                                  | 5% WB       | MCDBR | 11.1  | 11.6  | 12.6  | 12.5  | 12.4  | 11.2  | 10.5  | 10.2  | 11.2  | 10.8  | 10.7  | (38)  |      |
| (39) |                                                                                  | MCWBR       | 8.8   | 8.7   | 8.2   | 7.6   | 6.9   | 5.7   | 5.5   | 5.4   | 6.3   | 7.2   | 7.5   | 8.3   | (39)  |      |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.227 | 0.253 | 0.297 | 0.338 | 0.352 | 0.365 | 0.358 | 0.334 | 0.300 | 0.274 | 0.244 | 0.228 | (40)  |      |
| (41) |                                                                                  | taud        | 2.413 | 2.358 | 2.285 | 2.232 | 2.249 | 2.307 | 2.378 | 2.417 | 2.458 | 2.458 | 2.438 | 2.403 | (41)  |      |
| (42) |                                                                                  | Ebn at Noon | 908   | 938   | 932   | 917   | 913   | 899   | 901   | 912   | 921   | 902   | 873   | 863   | (42)  |      |
| (43) |                                                                                  | Edh at Noon | 67    | 88    | 111   | 130   | 132   | 126   | 116   | 107   | 93    | 79    | 65    | 60    | (43)  |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 2.02  | 3.12  | 4.59  | 5.89  | 6.74  | 6.67  | 6.32  | 5.73  | 4.94  | 3.46  | 2.20  | 1.65  | (44)  |      |
| (45) |                                                                                  | RadStd      | 0.13  | 0.15  | 0.17  | 0.22  | 0.34  | 0.39  | 0.32  | 0.34  | 0.21  | 0.14  | 0.14  | 0.10  | (45)  |      |

## Historical Trends

|      |                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only           | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (47) | Regional (6 neighbors) | N/A   | N/A     | -2.52  | N/A     | -0.78 | -0.87 | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page